

Figure 4.—Model of the hydrogen atom.

If we want to obtain a visual model of this atom, we must remember that the electron must move in a path around the nucleus, like a planet around the sun, for otherwise it would fall into the nucleus. We must therefore draw the path of the electron, and we shall assume for the moment, as a working hypothesis, that it is a circular orbit.

The hydrogen atom is the simplest of all atoms. The next one is the helium atom with its charge number 2—two elementary quanta of electricity on its nucleus and two planetary electrons in its extranuclear structure. And so forth, up to the heaviest atom known at the present time, curium, which has the nuclear charge number 96. All the necessary information on this subject is given in Table II, at the end of this book.

Once again, there arises the question, to what extent the visual representation supplied by Rutherford's atom model may be accepted literally, at its face value. Can we expect that some day, with the aid of a supermicroscope, we shall actually be able to see electrons revolving in their orbits around a nucleus? In view of the movements of the electrons, we would have to take snapshots. According to our present knowledge, we cannot very well doubt that a snapshot of a hydrogen atom would show actually such a picture as we have described: two point charges at a distance of a few ten-millionths of a millimetre. This is the practical significance of Rutherford's atom model. Obviously, this would not be a picture in definite colours, as our photographs would not be taken with rays of visible light, but with electron rays. But in any case such a snapshot of a hydrogen atom would show two particles: the nucleus and the single electron.

But if we were to develop the electron microscope into a moving picture camera of some sort, would it then be possible to follow the movement of the electron in its orbit around the nucleus, and to determine this orbit? Here we are faced with a fundamental difficulty, which makes it quite clearly evident that with this atom model we have reached the limits of visualization. For as soon as we have taken the first exposure on our film, we find that we no longer are in the position to take a second picture of the same atom. This is due to the fact that we will never again find it in an undisturbed condition. The atom has been disrupted by the very electrons which enabled us to take the first picture. The reason is that the first impact of the electrons used to